

OBSERVED STRESS-STRAIN BEHAVIOR OF CONFINED CONCRETE

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ABSTRACT: Thirty-one nearly full-size reinforced concrete columns, of circular, square, or rectangular wall cross section, and containing various arrangements of reinforcement, were loaded concentrically with axial compressive strain rates of up to 0.0167/s. The circular sections contained longitudinal and spiral reinforcement, the square sections contained longitudinal reinforcement and square and octagonal transverse hoops, and the rectangular wall sections contained longitudinal reinforcement and rectangular hoops with or without supplementary cross ties. The longitudinal stress-strain behavior of the confined concrete was measured and compared with that predicted by a previously derived stress-strain model with allows for the effects of various configurations of transverse confining reinforcement, cyclic loading, and strain rate. The measured longitudinal concrete compressive strain when the transverse steel first fractured was also compared with that predicted by equating the strain energy capacity of the transverse reinforcement to the strain energy stored in the concrete as a result of the confinement.

INTRODUCTION

In a companion paper by Mander et al. (1988), a theoretical stress-strain model for confined concrete was developed for members with either circular or rectangular sections, under static or dynamic axial compressive loading, either monotonically or cyclically applied. The concrete section may contain any general type of confinement with either spirals or circular hoops, or rectangular hoops with or without supplementary cross ties. For a particular transverse reinforcement configuration, the effective confining stresses f'_{lx} and f'_{ly} in the x and y directions can be calculated from the transverse reinforcement and the confinement effectiveness coefficient k_e , which defines the effectively confined concrete core area by taking into account the arching action that occurs between the transverse bars and between longitudinal bars. The form of the stress-strain curve for confined concrete was expressed in terms of three control parameters: the confined concrete compressive strength f'_{cc} , found using a constitutive model involving an ultimate strength surface for the applied axial stress and the confining stresses; the strain at the confined compressive strength, ϵ_{cc} ; and the modulus of elasticity of the concrete, E_c . The ultimate compressive concrete strain, ϵ_{cu} , defined as that strain at which first fracture of the

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transverse reinforcement occurred, was determined by equating the work done on the confined concrete and longitudinal reinforcement when deformed in compression to the available strain energy capacity of the transverse reinforcement when fracture occurred.

To extend the range of experimental results available to check the stress-strain theoretical model, tests were conducted on reinforced concrete short columns with either circular, square, or rectangular wall cross sections. The loading was applied concentrically at either quasi-static or high strain rates. Various arrangements of longitudinal and transverse reinforcement were investigated. The measured stress-strain behavior was compared with that predicted by the stress-strain model. This paper reports those experimental results and the comparison.

TESTS ON CONCENTRICALLY LOADED CIRCULAR COLUMNS CONFINED BY SPIRALS

General

Mander et al. (1984) conducted tests on cylinders of 500 mm (19.7 in.) diameter and 1,500 mm (59.1 in.) height. The cylinders were loaded concentrically in a DARTEC 10 MN (2,250 kips) servohydraulically controlled testing machine. Because of the high oil-pumping capacity, axial strain could be applied at strain rates of up to 0.015/s. The high strain rates imply that peak loads could be reached in as short a time as 0.2 s. Because of the servohydraulic nature of the machine control, the full falling branch behavior of the stress-strain relation could be monitored.

Details of the Columns

The details of the columns are given in Fig. 1 and Table 1. In the table, the symbols D and R used to describe reinforcement stand for deformed bar and round (plain) bar, respectively, and the following number is the bar diameter in millimeters (1 mm = 0.039 in.). Thus R12-52 means 12-mm-diameter round bar at 52 mm pitch; 12-D16 means 12 16-mm-diameter deformed bars.

It will be seen from Fig. 1 that the columns were constructed and tested in three series. A pilot series of three columns (a, b, and c) was followed by two series of six confined columns, each with a companion unreinforced column (CYL. 1 and CYL. 2) to enable the stress-strain curve of unconfined concrete to be assessed from tests on unreinforced units of the same size as the confined units so as to avoid scale effects. Column units in series 1 had identical longitudinal steel arrangements, but differing amounts and sizes of transverse spiral reinforcement, resulting in volumetric ratios of confining reinforcement, ρ_s , between 0.006 and 0.025. Series 2 column units had identical transverse reinforcement, but different amounts and sizes of longitudinal reinforcement. As shown in Fig. 1, the end 250 mm (9.9 in.) of each test unit had a reduced spiral spacing to ensure failure occurred in the central, instrumented region of the column. With the exception of column unit a, all column units were tested at the fast strain rate of 0.013/s. This strain rate was intended to simulate the effect of seismic loading rates of strain.

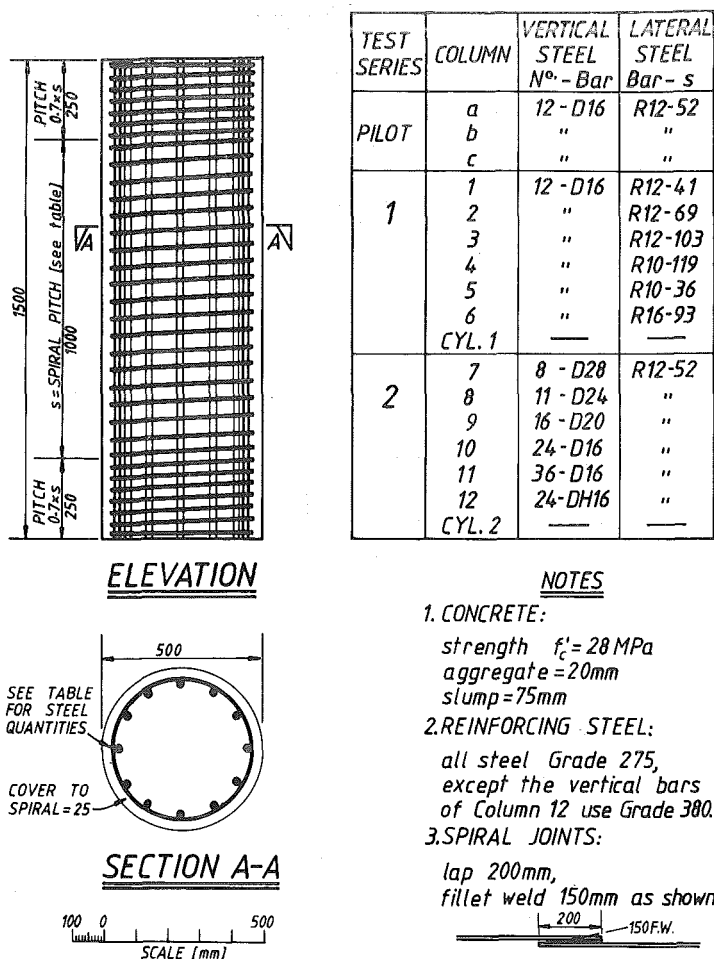


FIG. 1. Construction Details of Columns with Circular Cross Sections (1 mm = 0.0394 in.; 1 MPa = 145 psi)

Instrumentation

Monitoring of load applied by the DARTEC machine was available in the form of a visual digital display, and an analogue output. Longitudinal strains were recorded over the central 450-mm (17.7-in.) gage length of each column using four linear potentiometers at 90° intervals around the circumference measuring between steel rods integrally cast into and passing through the core. Clearance holes were provided in the cover concrete. It was assumed that strains indicated by these potentiometers were applicable both to the core concrete and to the longitudinal reinforcement, implying zero slip of longitudinal reinforcement.

Transverse strains on the spiral reinforcement were typically monitored at nine locations over the central 450-mm (17.7-in.) gage length, using

TABLE 1. Details of Spirally Reinforced Columns with Circular Cross Sections

Unit (1)	Longitudi- dinal steel ratio ^a	Longitudi- dinal steel ratio ^b	Trans- verse steel ratio	Material Strength ^c (MPa)			Core diam- eter (mm)	Confinement effective- ness ^d	Lateral confining pressure f_l^d (MPa)	Testing strain rate (s ⁻¹)
	ρ_l (2)	ρ_{cc} (3)	ρ_s (4)	f_c' (5)	f_y (6)	f_{yh} (7)	d_s (8)	k_e (9)		
a	0.0123	0.0160	0.020	28	295	310	438	0.970	3.0	0.000003
b	0.0123	0.0160	0.020	31	295	340	438	0.970	3.3	0.013
c	0.0123	0.0160	0.020	33	295	340	438	0.970	3.3	0.013 ^f
Cyl 1	—	—	—	28	—	—	—	—	—	0.013
1	0.0123	0.0160	0.025	28	295	340	438	0.983	4.18	0.013
2	0.0123	0.0160	0.015	28	295	340	438	0.950	2.42	0.013
3	0.0123	0.0160	0.010	28	295	340	438	0.911	1.55	0.013
4	0.0123	0.0159	0.006	28	295	320	440	0.890	0.85	0.013
5	0.0123	0.0159	0.020	28	295	320	440	0.986	3.14	0.013
6	0.0123	0.0163	0.020	28	295	307	434	0.926	2.84	0.013
Cyl 2	—	—	—	31	—	—	—	—	—	0.013
7	0.0251	0.0327	0.020	31	296	340	438	0.987	3.35	0.013
8	0.0253	0.0330	0.020	27	260	340	438	0.987	3.35	0.013
9	0.0256	0.0334	0.020	31	286	340	438	0.987	3.35	0.013
10	0.0246	0.0320	0.020	27	295	340	438	0.986	3.34	0.013
11	0.0369	0.0480	0.020	27	295	340	438	1.002 ^e	3.40	0.013
12	0.0246	0.0320	0.020	31	360	340	438	0.986	3.34	0.013

^aBased on gross section area.^bBased on core area.^cAt time of testing of units.^dFrom companion paper by Mander et al. (1988).^e K_e may exceed 1.0 by definition when ρ_{cc} is high.^fDynamic cyclic loading.

Note: 1 MPa = 145 psi; 1 mm = 0.039 in.

Kyowa KFC5-C1-11 electric resistance strain gages. All data were recorded in analog form on x-y plotters or chart recorders.

General Observations

Figs. 2(a-c) and 3(a-c) show column units 4 and 7, respectively, at three different stages of testing. In these photographs, it will be observed that two steel bands encircle the test units close to midheight. These in fact were separated from the unit by a 25-mm (1-in.) air gap all around, and had the purpose of protecting the potentiometers from damage as the cover concrete spalled.

Column unit 4, shown in Figs. 2(a-c) had the lowest volumetric ratio of confinement reinforcement ($\rho_s = 0.006$). Fig. 2(a) shows that just after peak load was registered, many vertical cracks had developed and crushing of the cover concrete was apparently more severe to the left of the cylinder. Fig. 2(b), which is just after the first hoop fracture, shows the development of a diagonal failure plane where the cover had spalled and was commencing to fall as large slabs of concrete. The full extent of the diagonal failure plane is shown in Fig. 2(c) at the end of the test when the instrumentation and loose concrete had been removed. This form of failure, with a strongly defined diagonal failure plane was characteristic of the test units with comparatively low volumetric ratios of confining steel.

Column unit 7, shown in Figs. 3(a-c), contained a moderately high amount of confinement reinforcement ($\rho_s = 0.020$). Fig. 3(a) shows column 7 with an axial strain of about 0.012 which was on the falling branch of the

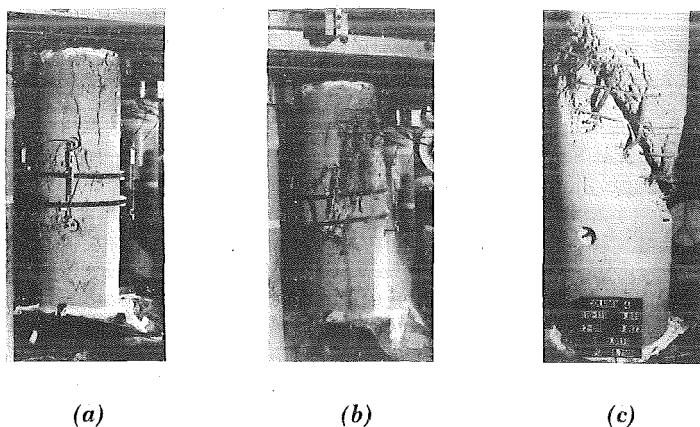


FIG. 2. Dynamic Testing of Circular Column with Low Spiral Volumetric Ratio (Column Unit 4; $\rho_s = 0.006$): (a) Just Following Peak Load at $\epsilon_c = 0.004$; (b) Just Following First Hoop Fracture at $\epsilon_c = 0.036$; and (c) End of Test

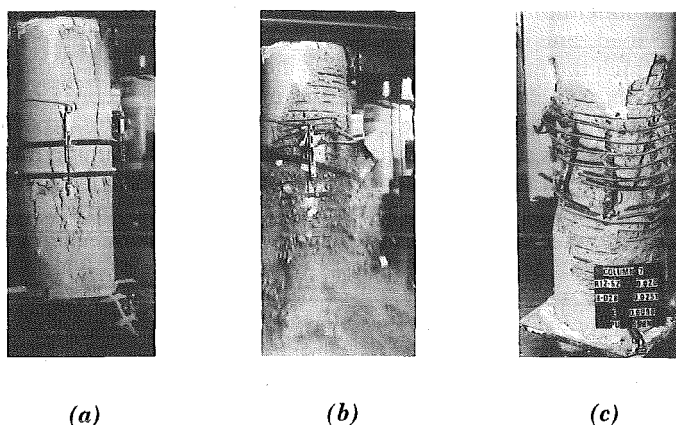


FIG. 3. Dynamic Testing of Circular Column with High Spiral Volumetric Ratio (Column Unit 7; $\rho_s = 0.020$): (a) Just Following Peak Load at $\epsilon_c = 0.012$; (b) Just Following Hoop Fracture at $\epsilon_c = 0.061$; and (c) End of Test

load-deformation response curve. Vertical cracks had developed in a symmetrical fashion around the central region of the unit. Fig. 3(b) shows the condition of the column just after the first hoop fracture. Deterioration of the load carrying capacity was rapid following the first hoop fracture, and the test was terminated with some six spiral fractures. The final photograph shows the condition at the end of the test, after loose cover and rubble had been removed. Note the fractured spirals, buckled longitudinal steel, and the lack of a well defined failure plane. This was characteristic of the column units with $\rho \geq 0.02$.

It is of interest that in no case did the spiral reinforcement tend to

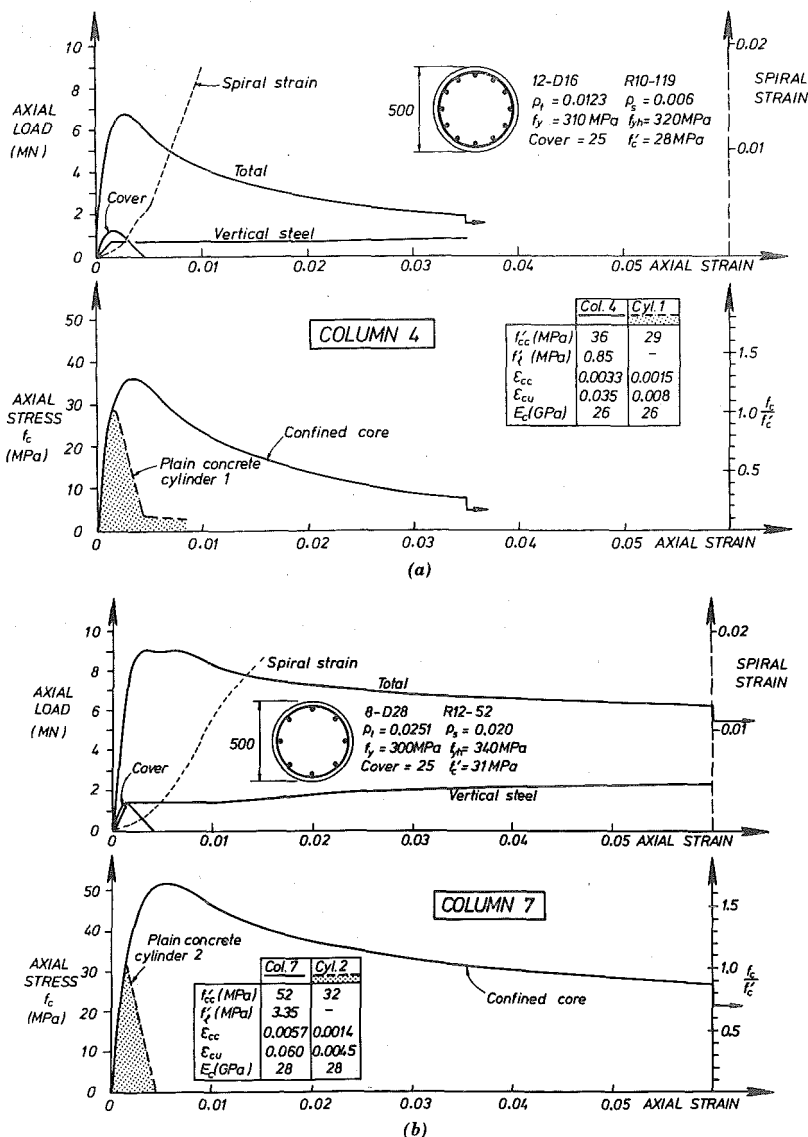
“unwind” after an initial hoop fracture. That is clearly illustrated in Fig. 2 at the end of the test where fractures had occurred sequentially over adjacent spirals.

Measured Stress-Strain Curves

Typical experimental stress-strain curves for confined concrete obtained from the test results are shown in Figs. 4(a) and 4(b) for columns with low confinement ratio (column 4) and high confinement ratio (column 7), respectively. As indicated previously, the columns were tested at a strain rate of 0.013/s. In the upper graph of each section, total load measured in the test, load carried by the vertical reinforcement determined from separate tests at identical rates of strain, and load carried by cover concrete determined from the comparison unreinforced column (CYL.1 or CYL.2) are separately plotted against axial (longitudinal) strain. The relationship between average spiral strain recorded by strain gages and axial strain is also included. The lower graph of each section plots axial stress in the confined core against axial strain, and it includes the measured stress-strain curve for the comparison unreinforced column. In these figures the core stress was found by subtracting the calculated loads carried by longitudinal reinforcement and cover concrete from the total measured load, and dividing by the core area. For both graphs, the axial strain at which the first spiral fractured is indicated by a step of decreasing load or stress, which occurs near the end of the strain record.

The method adopted to define axial strain, and used in the experimental stress-strain curves, requires some comment. Prior to attaining maximum load, the average of the four potentiometers around the circumference was used. Thereafter, between maximum load and first hoop fracture, the critical potentiometers were identified and used to determine increments of strain on the falling branch of the stress-strain curve. These critical potentiometers were identified by establishing that the failure region occurred wholly within the gage length of the potentiometers, and that the incremental displacement indicated by the critical potentiometers was approximately the same as the incremental stroke displacement. Thus, where the failure plane was steeply inclined, gage lengths that were not traversed by the failure plane were ignored. This is felt to be the most reasonable method for establishing the experimental strain in the failure region, as it is clear that the critical region for a column under combined bending and axial load will be well defined and will definitely contain the crushing region.

It can be seen from Fig. 4 that even for the comparatively lightly confined column 4, substantial enhancement of compression strength compared with unconfined concrete occurred, and a greatly extended falling branch resulted. For the more heavily confined column 7, the results are similar but more dramatic. For this column, first fracture of the spiral steel did not occur until an axial strain of 0.06 was attained. At this strain, the core concrete was still sustaining a stress in excess of $0.85f'_c$. It will be noted that the less heavily confined column 4 suffered hoop fracture at a much lower strain than column 7. This trend is apparent in the measured spiral steel strains, which rise more rapidly for column 4 than for column 7.



Comparison of Theory and Experiment

Figs. 5(a-d) and Table 2 compare the theoretical behavior predicted by the stress-strain model described in the companion paper by Mander et al. (1988), and the experimental behavior measured in the tests for the circular column units, and display the significance of different variables on the stress-strain behavior. From Figs. 5(a-d) it is apparent that the most significant variable in the tests was the volumetric ratio, ρ_s , of spiral reinforcement (Fig. 5[a]). Increasing ρ_s increases the peak stress f'_{cc} , the strain at peak stress ϵ'_{cc} , and the fracture strain ϵ_{cu} , and decreases the slope of the falling branch of the stress-strain curve. The effects of spiral pitch is shown in Fig. 5(b) for two columns with equal volumetric ratio of spiral reinforcement formed by either R10-36 centers or R16-93 centers. Although the 93-mm (3.7-in.) spacing gave a slightly less satisfactory falling branch than the 36-mm (1.4-in.) spacing, the influence is small. Peak stress and failure strain were not influenced significantly. Note that the larger spacing of 93 mm (3.7 in.) was 19% of the column diameter, which was close enough to effectively confine the concrete.

The number of longitudinal bars (Fig. 5[c]) has little influence on behavior, and no trend is obvious. Although considerable scatter occurs between test units with different amounts of longitudinal reinforcement (Fig. 5[d]), again no trend is obvious, and it is felt that the variations represent normal, though perhaps rather severe, experimental scatter.

The theoretical curves plotted for comparison in Figs. 5(a-d) were calculated using the experimental values for f'_{co} and ϵ'_{co} obtained from the comparison unreinforced column and the calculated effective lateral confining stress f'_l corresponding to yield of the spiral reinforcement (see Table 1). Details of the required theoretical parameters are included in Tables 1 and 2.

It will be seen that the theoretical curves in Figs. 5(a-d) give a good prediction of the full stress-strain curves. Also, estimates of the axial strain corresponding to first hoop fracture based on the energy balance method are in good average agreement with the experimental values.

In Table 2, the peak experimental and theoretical confined concrete strengths f'_{cc} are listed and compared. Agreement is very close in most cases with experimental concrete strengths exceeding the predicted level by 1.7% on average. The experimental confined strain at peak stress ϵ_{cc} is 1.3% less than the predicted confined strain on average. The experimental strain at first hoop fracture exceeded the predicted strain by 9.5% on average. These close agreements of average behavior, and the comparatively low scatter, indicate that the theoretical model presented in the companion paper gives an excellent prediction of the stress-strain curves for concrete confined by spirals.

Effect of Grade of Transverse Reinforcement

Typical stress-strain curves measured for New Zealand manufactured grade 275 ($f_{yh} \geq 40$ ksi) and grade 380 ($f_{yh} \geq 55$ ksi) are shown in Fig. 6. It is observed that the fracture strain of the high-strength steel is about 40% lower than that of the medium-strength steel. Also, the high-strength steel shows a greater increase in stress with increasing strain in the strain-hardening range after yielding commences. However, as mentioned in the companion paper, it has been found that the strain energy stored per unit

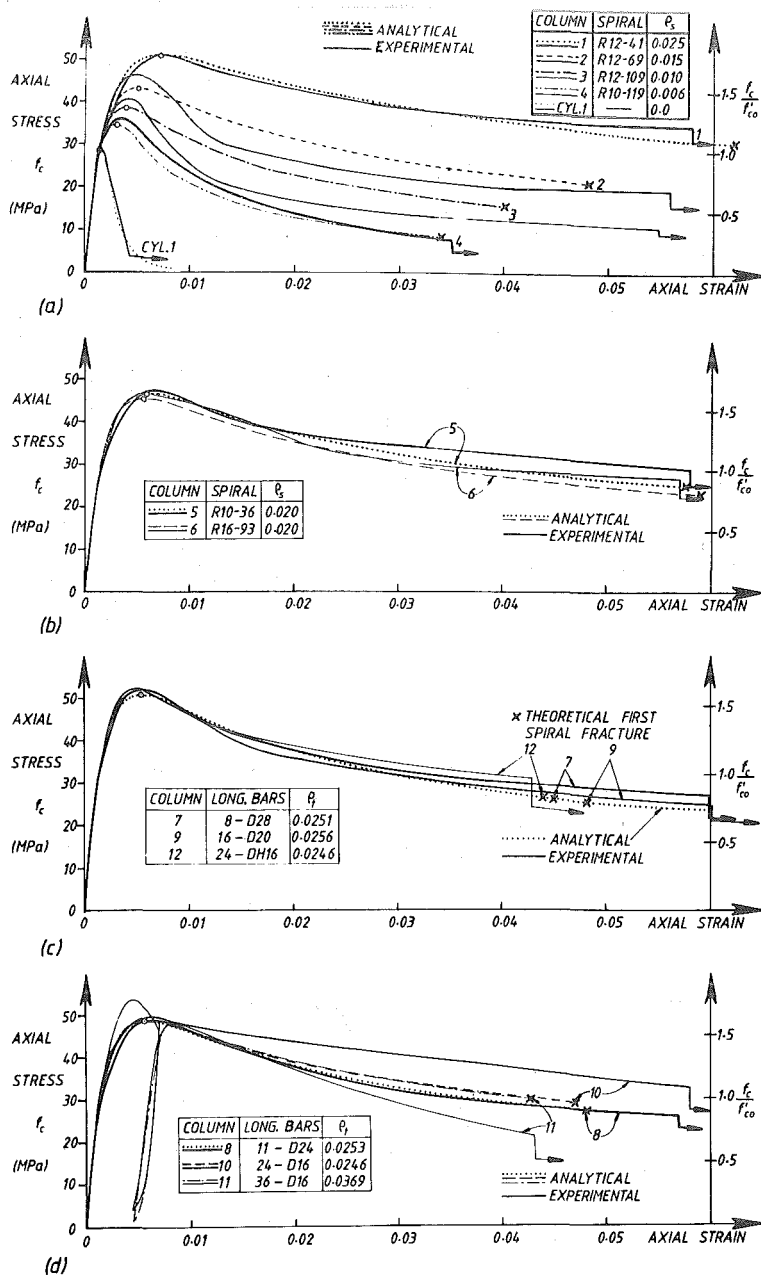


FIG. 5. Influence of Variables on Experimental Stress-Strain Curves and Comparison with Theory for Spirally Reinforced Circular Columns (1 MPa = 145 psi), Showing Effects of: (a) Amount of Spiral Reinforcement; (b) Spiral Pitch; (c) Number of Longitudinal Bars; and (d) Amount of Longitudinal Steel

TABLE 2. Experimental Results and Comparison with Theory for Spirally Reinforced Columns with Circular Cross Sections

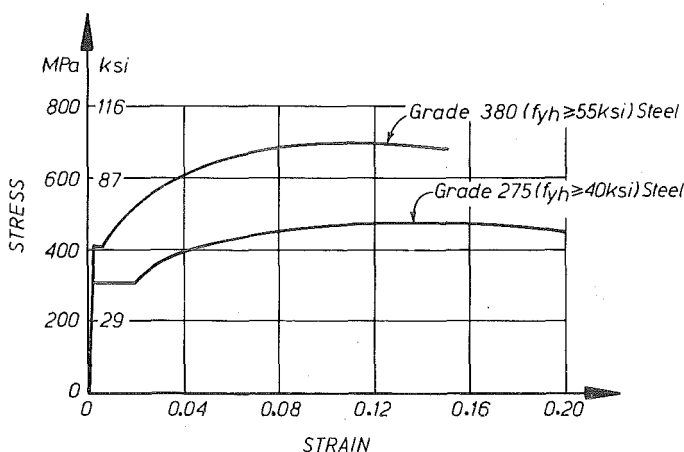
Unit (1)	Plain Concrete Data				Confined Strength f'_{cc}			Confined Strain ϵ_{rc}			Strain at Hoop Fracture ϵ_{ru}		
	f'_c (MPa) (2)	f'_{co} (MPa) (3)	ϵ_{co} (4)	E_c (GPa) (5)	Experimental (MPa) (6)	Theoretical ^a (MPa) (7)	Experimental/ Theoretical ^b (8)	Experimental (9)	Theoretical ^d (10)	Experimental/ Theoretical ^c (11)	Experimental (12)	Theoretical ^a (13)	Experimental/ Theoretical ^d (14)
a	30	24	0.002	24	38	40.3	0.94	0.008	0.0088	0.91	0.060	0.056	1.07
b	31	30	0.0015	31	48	48.3	0.99	0.0042	0.0061	0.69	0.039	0.053	0.74
c	33	32	0.0015	32	47	50.5	0.93	0.0058	0.0059	0.98	0.058	0.053	1.09
Cyl 1	28	29	0.0015	26	—	—	—	—	—	—	—	—	—
1	28	29	0.0015	26	51	51.0	1.00	0.0073	0.0072	1.01	0.058	0.062	0.94
2	28	29	0.0015	26	46	43.0	1.07	0.0050	0.0051	0.98	0.056	0.048	1.17
3	28	29	0.0015	26	40	38.5	1.04	0.0040	0.0039	1.03	0.055	0.040	1.38
4	28	29	0.0015	26	36	34.5	1.04	0.0033	0.0029	1.14	0.035	0.034	1.03
5	28	29	0.0015	26	47	46.5	1.01	0.0065	0.0060	1.08	0.058	0.057	1.02
6	28	29	0.0015	26	46	45.1	1.02	0.0058	0.0057	1.02	0.057	0.059	0.97
Cyl 2	31	32	0.0014	28	—	—	—	—	—	—	—	—	—
7	31	32	0.0014	28	52	50.8	1.02	0.0057	0.0055	1.04	0.060	0.045	1.33
8	27	30	0.0014	28	49	48.6	1.01	0.0058	0.0057	1.02	0.057	0.048	1.19
9	31	32	0.0014	28	52	50.8	1.02	0.0054	0.0055	0.98	0.060	0.048	1.25
10	27	30	0.0014	30	50	48.5	1.03	0.0064	0.0057	1.12	0.058	0.047	1.23
11	27	30	0.0014	30	54	48.8	1.11	0.0045	0.0057	0.79	0.0432	0.042	1.03
12	31	32	0.0014	28	52	50.7	1.03	0.0056	0.0055	1.02	0.043	0.044	0.98

^aTheoretical values computed from equations given in companion paper by Mander et al. (1988).^bAverage = 1.017.^cAverage = 0.987.^dAverage = 1.095.

Note: 1 MPa = 145 psi; 1 GPa = 145 ksi.

volume of reinforcing steel at fracture is approximately the same for both grades of steel. That is, the areas under the two stress-strain curves of Fig. 6 up to the fracture strains are approximately identical.

Equations for the required quantity of transverse reinforcement speci-

**FIG. 6. Typical Stress-Strain Curves for Steel Reinforcement (1 MPa = 145 psi)**

ed in design codes permit a reduction in the required quantity of transverse confining for higher strength steel. That is, the required quantity of transverse steel is inversely proportional to the yield strength of that steel, f_{yh} . However, it is possible that the available ductility of a section confined with high-strength transverse steel may be reduced due to premature fracture of that steel and a consequent loss of effective confinement. To investigate this possible problem six concentrically loaded spirally reinforced concrete columns were tested by Zahn et al. (1986). It was found that the volumetric ratio of confining reinforcement can be decreased with increasing steel yield strength without resulting in a reduction in the compressive strength of the confined concrete, and that the ductility of the confined concrete although reduced will remain high.

This result is explained by the energy balance approach described in the companion paper. The reduced volume of high-strength steel means a smaller strain energy capacity at fracture of that steel and hence a reduced ultimate concrete compressive strain ϵ_{cu} at first fracture of the transverse steel. Thus the ϵ_{cu} available using grade 380 transverse steel will be approximately $275/380 = 0.72$ times the ϵ_{cu} available using grade 275 transverse steel.

TESTS ON CONCENTRICALLY LOADED SQUARE COLUMNS CONFINED BY SQUARE AND OCTAGONAL HOOPS

General

Scott et al. (1982) have reported a test program involving 30 column units of 450 mm $\times$ 450 mm (17.7 in. $\times$ 17.7 in.) square cross section and 1,200 mm (47.2 in.) in height. The column units were loaded concentrically in the DARTEC testing machine in a fashion similar to the circular columns.

Effect of Age at Testing

Five column units constructed but not tested by Scott et al. (1982) were tested under concentric compression at an average age of greater than 940 days. These tests were made to check whether any ageing effects are apparent apart from strength increase and are fully described elsewhere by Mander et al. (1984).

Figs. 7(a-c) and 8(a-c) show experimental results for columns with 12 longitudinal bars and 8 longitudinal bars, respectively, and with different hoop arrangements. The construction details of the columns are also shown in figures. In Figs. 7(a-c) the results of column 13 tested by Scott et al. (1982) at age 67 days are compared with the results of column 16 tested at age 942 days. In Figs. 8(a-c), the results of columns 6 and 7 tested by Scott et al. at ages 75 and 76 days, respectively, are compared with the results of column 10 tested at age 983 days. For all columns, the strain rate was $\epsilon_c = 0.0167/s$ except for column 6 where it was $0.0000033/s$. The axial strains were recorded over the central 400-mm (15.7-in.) gage length of column using four linear potentiometers, one at each face of the column measuring between steel rods as for the circular columns. In these figures, the axial strain plotted is defined as for the circular column tests, namely the average strain occurring in those gage lengths within the crushing region rather than the average strain of all gage lengths. The experimental

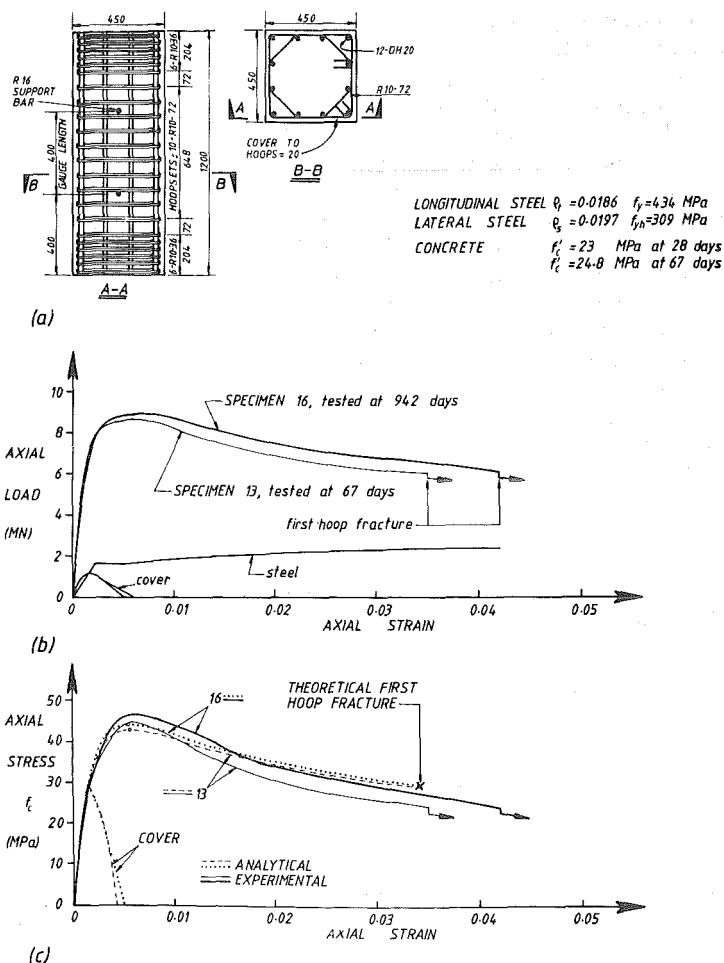
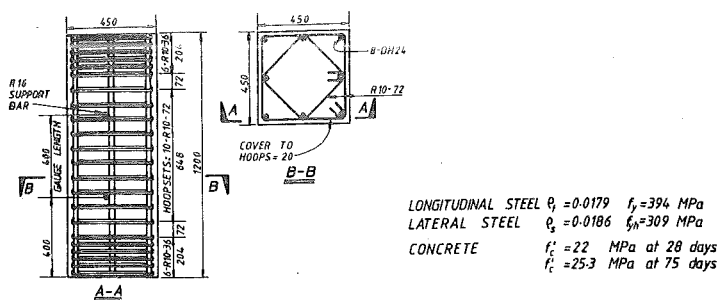


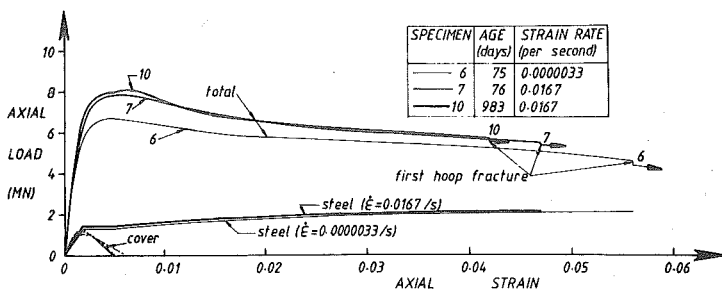
FIG. 7. Influence of Age at Testing on Stress-Strain Curves for 12-Bar Square Columns (1 MPa = 145 psi; 1 mm = 0.0394 in.): (a) Details of Specimens 13 and 16; (b) Specimen Load-Strain Response; and (c) Concrete Stress-Strain Response

stress-strain curves for the core concrete were found by subtracting the load carried by the cover concrete and the longitudinal reinforcement from the total load, and dividing by the core area.

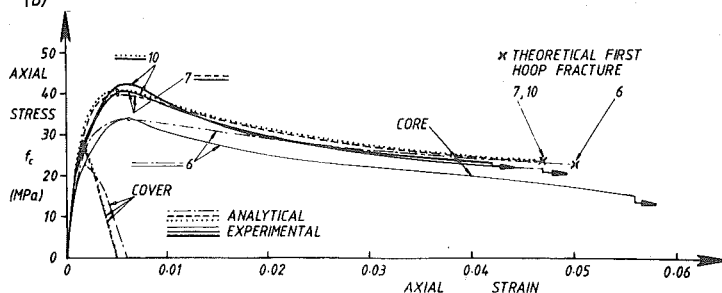
From Figs. 7(a-c), it will be seen that the 12-bar column 16 tested at 942 days attained a slightly higher stress and ultimate strain than column 13 tested at 67 days. However, from Figs. 8(a-c), it is evident that the eight-bar column 10 tested at 983 days produced an almost identical stress-strain curve to column 7 which was tested at 76 days, except for a reduction in ultimate strain for column 10. Column 6, tested at the quasi-static strain rate at age 75 days, attained lower strength but higher strain at first hoop fracture. These figures indicate that apart from the slight



(a)



(b)



(c)

FIG. 8. Influence of Age at Testing on Stress-Strain Curves for Eight-Bar Square Columns (1 MPa = 145 psi; 1 mm = 0.0394 in.): (a) Details of Specimens 6, 7, and 10; (b) Specimen Load-Strain Response; and (c) Concrete Stress-Strain Response

strength increase expected due to the increased age of the concrete at time of testing, no significant ageing effects are apparent.

Comparison between theoretical stress-strain curves predicted by the proposed model and the experimental results, are included for columns 13 and 16 in Figs. 7(a-c) and for columns 6, 7, and 10 in Figs. 8(a-c). Good agreement is apparent for the shape of the curves as well as peak strains at first hoop fracture.

TESTS ON CONCENTRICALLY LOADED RECTANGULAR WALL SECTIONS CONFINED BY RECTANGULAR HOOPS AND CROSS TIES

General

Unlike the stress-strain models proposed by previous investigators, the theoretical model proposed in the companion paper is able to cope with

rectangular sections with different levels of confinement in the two principal directions. The concern of Mander et al. (1984) when developing the model was particularly with the behavior of flanges of hollow rectangular box-section columns under seismic response, although clearly other situations have relevance, such as solid rectangular walls or slab piers. Tests were conducted on sixteen rectangular wall units to observe behavior and to check the accuracy of the stress-strain model.

Details of the Units

Fig. 9 presents the details of 16 rectangular wall units. Each wall had a cross section of 150 mm (5.9 in.) $\times$ 700 mm (27.6 in.) with an overall height of 1,200 mm (47.2 in.). The test units were designed as models of one-half of full size of the flange region of a 1,900 mm (74.8 in.) $\times$ 1,900 mm (74.8 in.) square hollow columns with 300 mm (11.8 in.) thick walls. A cover of 25 mm (1 in.) to the outside hoop steel was adopted.

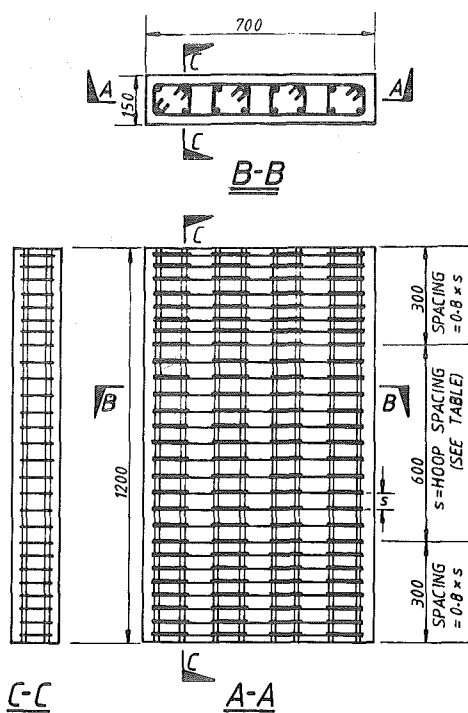
The principal variables in this test series were the amount and configuration of the transverse reinforcement used in the wall units. Three different hoop configurations were examined, including square hoops enclosing four adjacent longitudinal bars together with an external perimeter hoop (walls 1, 2, 5, 6, and 9–12), overlapping square hoops together with an external perimeter hoop (wall 3), and transverse cross ties together with an external perimeter hoop (walls 4, 13, and 14). The amount of transverse hoop steel chosen covered the typical range used in columns designed in accordance with the New Zealand concrete design code *NZS 3101* (Standards Association of New Zealand 1982).

It should be noted that several of the units violated the spacing requirements of *NZS 3101*. The requirement that the spacing of hoop sets should not exceed 0.2 of the least lateral dimension could be very onerous for hollow section columns if the least lateral dimensions is taken as the wall thickness, and would result in a maximum spacing of $150/5 = 30$ mm (1.2 in.) for the hoop sets in these wall units (i.e., 60 mm [2.4 in.] in prototype dimension). Walls 3, 5, 6, 11, and 12 deliberately violated this requirement. Also, in prototype terms, walls 13 and 14 had a center to center spacing of longitudinal reinforcement of 320 mm (12.6 in.) which exceeds the maximum of 200 mm (7.9 in.) permitted by *NZS 3101*. Nevertheless, it will be shown later that these code violations did not result in unsatisfactory performance.

The wall units were cast in two batches of eight units each, two of which were unreinforced to allow the unconfined stress-strain curve to be examined at both high and low strain rates. Full details of the wall unit reinforcement, material strengths and confining stresses are given in Table 3. Of particular interest in this table are the very low confinement effectiveness coefficient values which vary in the range of $0.40 \leq k_e \leq 0.70$. These low values stem from the comparatively wide spacing of longitudinal and transverse reinforcement compared with the width of the confined core. Also of interest are the different levels of confining stress provided in the two principal transverse directions.

Instrumentation

Longitudinal strains were measured by four potentiometers recording over the central 400-mm (15.7-in.) gage length of each wall unit. Mounting



NOTES

1. STEEL:
All steel GRADE 275
2. CONCRETE:
Strength $f'_c = 28\text{MPa}$
Aggregate = 13mm
Slump = 150mm
3. COVER:
Cover to hoops = 25mm.

STEEL DETAILS FOR WALLS 1,2,5,6, and 9 to 12.



WALL 3



WALL 4



WALLS 13 and 14

BATCH	WALL	VERTICAL STEEL N ^o — Bar	LATERAL STEEL Bar — s
1	1	16-D12	R6-25
	2	"	"
	3	"	R6-50
	4	"	R6-25
	5	"	R6-72
	6	"	R10-42
	7	—	—
	8	—	—
2	9	16-D12	R6-25
	10	16-D16	"
	11	"	R6-50
	12	"	R10-42
	13	10-D12	R10-30
	14	"	R6-30
	15	—	—
	16	—	—

FIG. 9. Construction Details of Columns with Rectangular Sections (1 MPa = 145 psi; 1 mm = 0.0394 in.)

TABLE 3. Details of Columns with Rectangular Cross Sections

Wall (1)	Vertical Reinforcement				Transverse Reinforcement ^d				Confining Stress			
	$f_c^{'a}$ (MPa) (2)	ρ_l^b (3)	ρ_{cc}^c (4)	f_y (MPa) (5)	ρ_y^e (6)	ρ_x^e (7)	$\rho_s = \rho_x + \rho_y$ (8)	f_{yh} (9)	k_e^f (10)	f_{ly}^f (MPa) (11)	f_{lx}^f (MPa) (12)	Test strain rate (s ⁻¹) (13)
1	28	0.0172	0.030	330	0.0196	0.0270	0.0466	310	0.650	3.97	5.44	0.00001
2	28	0.0172	0.030	330	0.0196	0.0270	0.0466	340	0.650	4.33	5.97	0.0133
3	28	0.0172	0.030	330	0.0157	0.0270	0.0427	310	0.541	2.64	4.52	0.00001
4	28	0.0172	0.030	330	0.0157	0.0270	0.0427	310	0.650	3.16	5.44	0.00001
5	28	0.0172	0.030	330	0.0068	0.0094	0.0162	310	0.454	0.95	1.32	0.00001
6	28	0.0172	0.031	330	0.0292	0.0416	0.0708	330	0.587	5.66	8.06	0.00001
7	28	—	—	—	—	—	—	—	—	—	—	0.0133
8	28	—	—	—	—	—	—	—	—	—	—	0.00001
9	41	0.0172	0.030	330	0.0196	0.0270	0.0466	340	0.650	4.35	5.97	0.0133
10	41	0.0306	0.053	290	0.0196	0.0270	0.0466	340	0.700	4.69	6.43	0.0133
11	41	0.0306	0.053	290	0.0098	0.0135	0.0233	340	0.586	1.95	2.69	0.0133
12	41	0.0306	0.056	290	0.0292	0.0416	0.0708	360	0.634	6.66	9.49	0.0133
13	41	0.0108	0.020	290	0.0205	0.0582	0.0787	360	0.399	2.94	8.36	0.0133
14	41	0.0108	0.019	290	0.0082	0.0225	0.0307	340	0.397	1.11	3.04	0.0133
15	41	—	—	—	—	—	—	—	—	—	—	0.0133
16	41	—	—	—	—	—	—	—	—	—	—	0.00001

^aAverage age at testing: approximately 240 days.^bLongitudinal steel ratio related to gross section area.^cLongitudinal steel ratio related to core section area.^dSee Fig. 9 for configuration of transverse reinforcement.^ey = section short direction; x = section long direction.^fGiven by the equations in companion paper by Mander et al. (1988).

Note: 1 MPa = 145 psi.

rods for the potentiometers were located 175 mm (6.9 in.) in from each end face of the units passing through the core concrete. Transverse strains were measured by electric resistance strain gage on hoop reinforcement at the central cross section.

Testing Procedure

Because of the rather high aspect ratio of $1,200/150 = 8$, the wall units possessed a degree of lateral instability not apparent in the prototype box-columns, where lateral support for the compression flange is provided by the webs. This instability caused the first wall unit tested (wall 1) to fail by buckling before fracture of hoop reinforcement occurred. To avoid further instability, a stabilizing frame was built which provided lateral support to the edges of the wall units, but did not inhibit longitudinal strain, or any tendency to buckle midway between the lines of lateral support. Thus the stabilizing effects of webs was effectively modelled for all but wall 1.

The walls were loaded concentrically. Most wall units were loaded monotonically at either the quasi-static loading rate ($\epsilon_c = 0.00001/s$) or the high strain rate ($\epsilon_c = 0.0133$). Three wall units, however, were unloaded and reloaded once or more during the test program to investigate the influence of repeated loading on the stress-strain curves.

Results

Experimental stress-strain curves for the unreinforced concrete wall units are plotted in Fig. 10, and show the influence of dynamic loading. This figure also presents the experimental values for the strength pa-

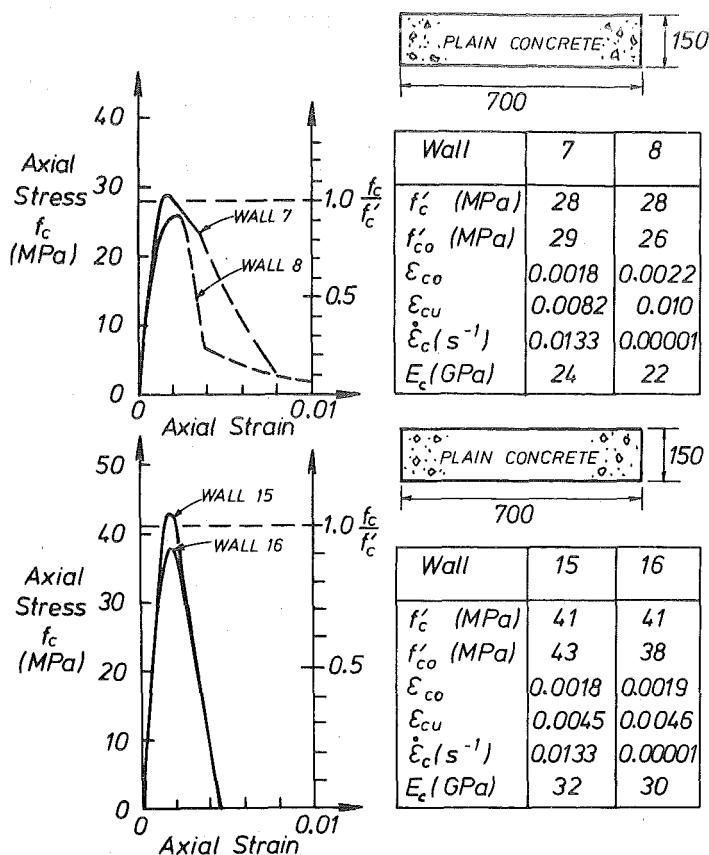


FIG. 10. Experimental Stress-Strain Curves for Unreinforced Columns with Rectangular Sections (1 MPa = 145 psi; 1 mm = 0.0394 in.)

rameters f'_{co} and ϵ_{co} , and the cylinder compressive strength f'_c . The experimental values of f'_{co} and ϵ_{co} were used to predict the stress-strain response of the reinforced wall units using the theoretical model described in the companion paper.

Experimental results for axial load versus axial strain, and core stress versus axial strain are shown in Fig. 11 for a typical test. The axial load-strain plot shows a pronounced drop at an axial strain of approximately 0.003, corresponding to spalling of cover concrete. Strength enhancement of the confined core subsequently results in an increase in load to a level higher than that attained prior to spalling. The wall unit shown in Fig. 11 was unloaded and reloaded twice during the test program. It should be noted that the unloading occurs at a stiffness similar to the initial elastic stiffness, and that the stress-strain curve appears to regain the monotonic level. As with other tests previously described, the core stress is found by subtracting the load carried by cover concrete and longitudinal

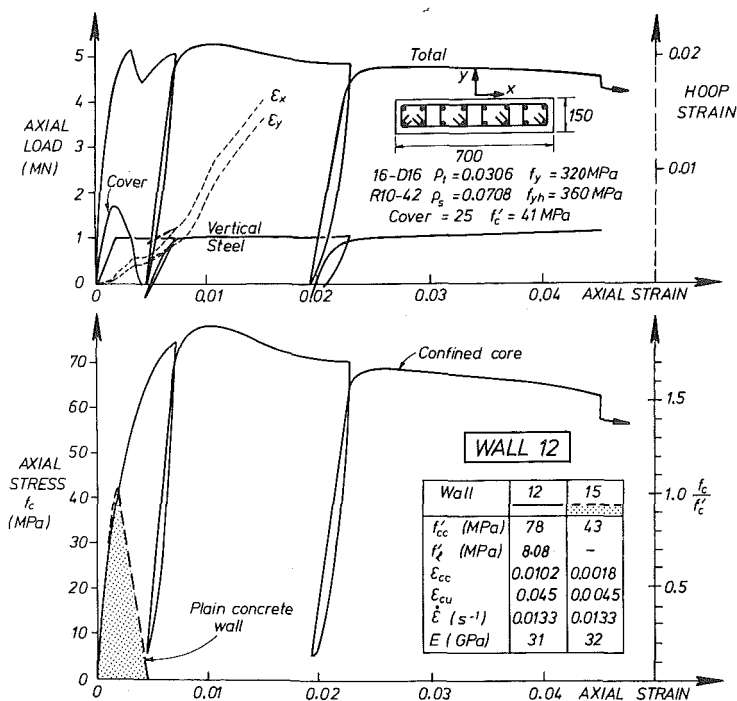


FIG. 11. Typical Experimental Stress-Strain Curves for Columns with Rectangular Sections (1 MPa = 145 psi; 1 mm = 0.0394 in.)

reinforcement from the total load, and dividing by the core area (measured to the center line of the perimeter hoop reinforcement).

Figs. 12(a-d) and Table 4 summarize the experimental results and compare with the predicted theoretical values. The effect of different arrangements of transverse hoop reinforcement on the stress-strain behavior is shown in Fig. 12(a). All three wall units shown contained the same amount of transverse reinforcement, and were tested at the same strain-rate of $\epsilon_c = 0.00001/s$. Of the three wall units, wall 4, with cross ties, gave the best performance. The first hoop fracture in wall 3 with overlapping square hoops occurred at the rather low strain of $\epsilon_{cu} = 0.028$, and is believed to be due to nonuniform action of the two overlapping hoops bent around each longitudinal bar. Although wall 1, with nonoverlapping hoops also gave poor response, this was due to premature wall buckling, as discussed earlier. It is believed that had this wall been adequately supported at its edges, as were walls 3 and 4, its behavior would also have been satisfactory. This belief is supported by the behavior of wall 2, which was identical to wall 1 except that it was loaded at the high strain rate (see Fig. 12[c]), and behaved satisfactorily. Despite the problems noted for walls 1 and 3 relating to ultimate strain, the theoretical predictions of the general stress-strain curve shapes, shown in Fig. 12(a), are good. It will be noted that despite nominally identical confinement ratios and concrete strength, the theoretical predictions for the three hoop configurations

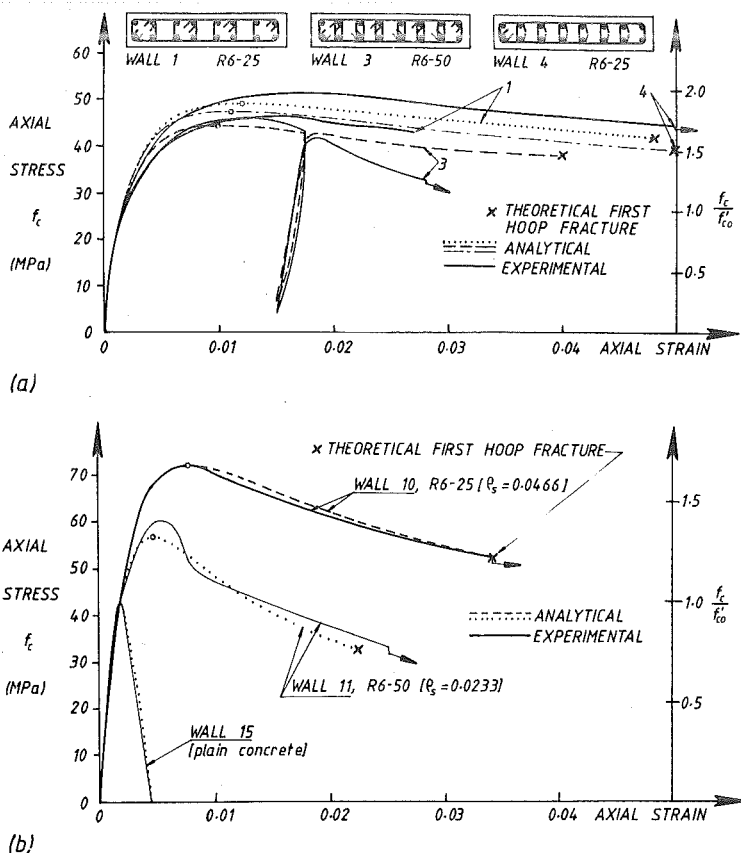


FIG. 12. Influence of Variables on Experimental Stress-Strain Curves and Comparison with Theory for Columns with Rectangular Sections (1 MPa = 145 psi; 1 mm = 0.0394 in.), Showing Effects of: (a) Hoop Configuration; (b) Hoop Spacing with Constant Hoop Size; (c) Strain Rate; and (d) Longitudinal Steel Distribution

differ. This is primarily a result of the different transverse steel configurations for these walls.

Fig. 12(b) compares the influence of amount of confining reinforcement on the stress-strain curves, and includes a comparison with unconfined concrete. The trends apparent are identical to those reported earlier for circular-section and rectangular-wall units. It will be noted that the theoretical predictions for all three stress-strain curves in this figure are extremely close to the experimental data.

The influence of strain rate on the stress-strain curve is illustrated in Fig. 12(c). Within the limits of comparison imposed by the premature buckling of wall 1, the trends conform to those for other sections reported earlier, with an increase in strength and slope of the falling branch resulting from an increased strain rate.

The effect of altering the distribution of longitudinal reinforcement is shown in Fig. 12(d). Walls 12 and 13 had similar amounts of confining

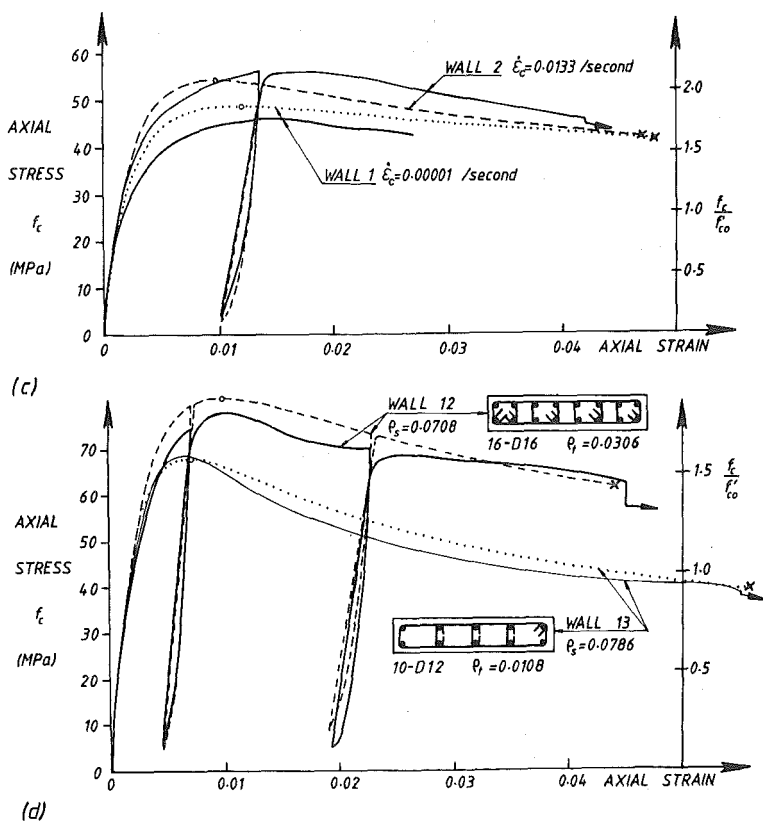


FIG. 12. continued

reinforcement ($\rho_s = 0.0708$ and 0.0786 , respectively) but it can be seen that the stress-strain response of the two units is quite different. Wall 12 had 16 closely spaced 16-mm- (0.63-in.-) diameter bars ($\rho_t = 0.0306$), while wall 13 had 12 more widely spaced 12-mm- (0.47-in.-) diameter bars ($\rho_t = 0.0108$). According to the theoretical model presented in the companion paper, the confinement effectiveness coefficients k_e were 0.63 and 0.40 for walls 12 and 13, respectively. Thus it is evident that a closer spacing of longitudinal bars and cross ties will lead to more effectively confined concrete with a greater strength enhancement. Nevertheless, it should be noted that by using extra longitudinal steel to improve the axial load carrying capacity, additional strain energy input from the hoop reinforcement is required to restrain the longitudinal bars from buckling. Hence for a constant volume of confining steel, the strain at first hoop fracture will be attained sooner for sections in which the longitudinal steel has a large load carrying capacity.

It should be noted that the difference in behavior apparent in Fig. 12(d) does not result from the different longitudinal reinforcement configurations, comparisons of walls with similar longitudinal reinforcement configurations,

TABLE 4. Experimental Results and Comparison with Theory for Columns with Rectangular Cross Sections

Wall (1)	Plain Concrete Data ^a			Confined Strength f'_{cc}			Confined Strain ϵ_{cu}			Strain at Hoop Fracture ϵ_{cu}		
	f'_{co} (MPa) (2)	ϵ_{co} (3)	E_c (GPa) (4)	Experi- mental (MPa) (5)	Theo- retical ^b (MPa) (6)	Experi- mental/ Theo- retical (7)	Experi- mental (8)	Theo- retical ^b (9)	Experi- mental/ Theo- retical (10)	Experi- mental (11)	Theo- retical ^b (12)	Experi- mental/ Theo- retical (13)
1	26	0.0022	22	46	49	0.94	0.014	0.012	1.17	0.027 ^c	0.048	0.56 ^c
2	29	0.0018	24	56	55	1.02	0.017	0.0098	1.73	0.042	0.047	0.89
3	26	0.0022	22	46	44	1.05	0.013	0.010	1.30	0.028	0.040	0.70
4	26	0.0022	22	51	47	1.08	0.019	0.0111	1.71	0.050	0.050	1.00
5	26	0.0022	22	37	33	1.12	0.0075	0.0052	1.44	0.0284	0.025	1.14
6	26	0.0022	22	56	56	1.00	0.021	0.015	1.40	0.055	0.062	0.89
9	43	0.0018	32	72	71	1.01	0.0082	0.0076	1.08	0.030	0.040	0.75
10	43	0.0018	32	72	72	1.00	0.0076	0.0079	0.96	0.034	0.034	1.00
11	43	0.0018	32	60	57	1.05	0.0052	0.0047	1.11	0.025	0.0225	1.11
12	43	0.0018	32	78	81	0.96	0.0102	0.0098	1.04	0.045	0.044	1.02
13	43	0.0018	32	69	69	1.00	0.0065	0.0072	0.90	0.055	0.056	0.98
14	43	0.0018	32	58	55	1.05	0.0046	0.0043	1.70	0.033	0.032	1.03

^aFrom tests on walls 7, 8, 15, and 16 (see Fig. 10).^bFrom equations given in the companion paper by Mander et al. (1988).^cHoop fracture did not occur: wall failed prematurely by buckling.

Note: 1 MPa = 145 psi; 1 GPa = 145 ksi.

but different bar diameter (for example, walls 9 and 10) indicated almost identical stress-strain curves to those shown here (see Table 4 for salient details).

Figs. 12(a–d) also illustrate that the theoretical stress-strain model proposed in the companion paper for cyclic loading gives good agreement with the unloading and loading branches in those cases where the loading was cycled.

Predicted and experimentally obtained values for the confined strength f'_{cc} , the confined strain at peak stress ϵ_{cc} , and the strain at first hoop fracture ϵ_{cu} are listed in Table 4, together with the relevant ratio of experimental to theoretical results. It will be noted that very good agreement is obtained for the confined strength, with the theoretical predictions tending to be slightly conservative. Much more scatter is apparent in the strain at peak stress. For the slow strain rate tests, the theoretical predictions are very conservative, although with the exception of wall 2, the agreement is good for the high-strain rate results. With the exception of walls 1 and 3, for the reasons noted earlier, the agreement between ultimate strain based on the energy balance method measured strain at first hoop fracture is quite good.

CONCLUSIONS

Tests were conducted on reinforced concrete short columns with either circular, square, or rectangular cross sections. The loading was applied concentrically at either quasi-static or high strain rates. Various arrangements of longitudinal and transverse reinforcement were investigated. The following trends were observed:

1. The most significant parameter affecting the shape of the stress-strain curve of confined concrete for all section shapes was the quantity of

confining reinforcement, in the form of spirals for circular columns, or rectangular hoops or cross ties for square or rectangular columns. As the volumetric ratio of confining reinforcement increased, the strength developed increased, the slope of the falling branch decreased (i.e., flattened), and the longitudinal strain at which hoop fracture occurred increased. These trends followed the theoretical predictions of a stress-strain model proposed by the writers for confined concrete.

2. The influence of the configuration of transverse reinforcement can be predicted through the confinement effectiveness coefficient k_e . The configuration of transverse reinforcement had a particularly large effect, with k_e varying in the range 0.40–0.70 for the rectangular walls and 0.89–1.0 for the circular columns. Also the only significance of the configuration of longitudinal reinforcement was through its effect on k_e .

3. As predicted, the circular columns confined with spiral reinforcement performed better than rectangular or square columns. This was apparent in both the strength enhancement and the ultimate compression strain for a given volumetric confinement ratio.

4. The analytical stress-strain model proposed by the writers and described in the companion paper was found to give good prediction of experimental behavior for circular, square, and rectangular columns with various reinforcement configurations. As well as accurately predicting the enhanced strength and general shape of the stress-strain curves for the confined concrete, the longitudinal strain at first hoop fracture was predicted within surprisingly close tolerances using the energy balance approach. This approach, which relates the increase in strain energy absorbed by the confined core to the strain energy available in the yielding hoop sets provides a neat explanation for the enhancement of ductility of concrete confined by spirals or hoops.

ACKNOWLEDGMENTS

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APPENDIX II. NOTATION

The following symbols are used in this paper:

b_c	=	concrete core dimension to center line of perimeter hoop in x direction;
D_f	=	dynamic magnification factor for concrete strength due to dynamic loading;
d_c	=	concrete core dimension to center line of perimeter hoop in y direction;
d_s	=	diameter of spiral;
E_c	=	modulus of elasticity of concrete;
f_c	=	longitudinal concrete stress;
f'_c	=	compressive strength of unconfined concrete cylinder 150 mm (6 in.) diameter $\times$ 300 mm (12 in.) high;
f'_{cc}	=	compressive strength (peak stress) of confined concrete;
f'_{co}	=	compressive strength of unconfined concrete;
$(f'_{co})_{dyn}$	=	dynamic compressive strength of unconfined concrete;
f'_l	=	effective lateral confining stress;
f'_{lx}	=	effective lateral confining stress in x direction;
f'_{ly}	=	effective lateral confining stress in y direction;
f_y	=	yield stress of longitudinal reinforcing steel;
f_{yh}	=	yield strength of transverse reinforcing steel;
k_e	=	confinement effectiveness coefficient; and
s	=	spiral spacing pitch.